

AutoML and AutoRAG Strategies

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Home

Goal

Teach technical professionals how to strategically evaluate use cases and apply AutoML (predictive modeling) and AutoRAG (generative retrieval) in Red Hat OpenShift AI 3.5+ to drive measurable business value by showcasing its targeted automated AI capabilities.

Objectives

On completing this course, you should be able to:

- **Predictive Foundations (AutoML):** Evaluate when to apply AutoML to convert raw tabular data or time series into production-ready predictive models across classification, regression, and time-series tasks.
- **AutoML governance & deployment:** Use AutoML evaluation artifacts (leaderboards, feature importance, ROC/PR curves) to validate business logic and promote winning models to production.
- **GenAI prompting & MLflow governance:** Transition from predictive data to generative text, leveraging the GenAI Playground and MLflow to version, track, and eliminate "prompt drift".
- **Generative retrieval (AutoRAG):** Position AutoRAG as an automated optimization engine that benchmarks chunking, embedding, and vector database configurations for grounded LLM applications.
- **Knowledge review and assessment:** Track core enablement concepts across AutoML, MLflow prompt tracking, and AutoRAG to validate practitioner capability in diagnosing enterprise data scenarios and selecting optimal AI patterns.

Audience

- **Primary Personas:** Data Scientists, AI Engineers, and Technical Solution Architects evaluating or adopting Red Hat OpenShift AI 3.5+.
- **Technical Level & Focus (TL2):** Technical practitioners and field specialists looking to understand the "why", decision points, trade-offs, and business outcomes of automated capabilities. Focused on transitioning from manual ML/RAG development without needing to perform cluster admin setup procedures.
- **Distribution & Accessibility:** Publicly discoverable enablement designed for global Red Hat field teams, ecosystem partners, and enterprise customers.

Prerequisites

This course assumes that you have the following experience:

- Familiarity with core machine learning concepts (classification, regression, time series).

- Familiarity with generative AI concepts (LLMs, prompting, retrieval-augmented generation).
- Familiarity with Red Hat OpenShift AI 3.5+ from a practitioner (Data Scientist / AI Engineer) perspective.

Notice that no cluster administration or infrastructure setup experience is assumed or required.

Classroom environment

This course requires no hands-on user labs. It uses diagrams, Summit Arcade interactive clickthroughs, and video walkthroughs to demonstrate AutoML and AutoRAG capabilities.

Other sources of information about Red Hat OpenShift AI

For documentation about Red Hat OpenShift AI, see either its [product documentation](#) or the [Red Hat AI Examples](#) upstream repositories.

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Predictive Foundations with AutoML

Goal

Evaluate when to apply AutoML to convert raw tabular data or time series into production-ready predictive models across classification, regression, and time-series tasks.



Not started yet.

This chapter introduces AutoML as the fastest path from raw data to ranked predictive models, and helps you recognize which prediction task type fits each business problem.

From Raw Data to Ranked Models

Estimated reading time: 10 minutes.

Objective

Recognize the four AutoML prediction task types and the business economics that make automated model generation compelling.



Not started yet.

The business case for AutoML

At the C-Suite level, AutoML compresses model development from months to hours, delivering 70–80% faster time-to-market and acting as a 3x workforce multiplier. Think of AutoML as a "Sorting Hat" that converts raw CSVs pulled from S3 into a ranked leaderboard of candidate models.

Choosing a prediction task type

Every predictive problem maps to one of four task types:

- **Binary** — a "Yes/No" decision.
- **Multiclass** — choosing from a "menu" of categories.
- **Regression** — answering "How much?".
- **Time Series** — predicting the "next value over time".

Sizing economics

Predictive models are cheap to run: microsecond CPU inference with KB/MB-sized artifacts, in sharp contrast to GPU-hungry LLMs. Understanding this cost profile is key to justifying where AutoML fits in an enterprise portfolio.

What's Next

Now that you can match a business problem to the right AutoML task type, the next chapter opens the hood on how AutoML builds and stacks its models — and how to read the evaluation "report

card" before promoting a winner to production.

Validating and Deploying AutoML Models

Goal

Use AutoML evaluation artifacts (leaderboards, feature importance, ROC/PR curves) to validate business logic and promote winning models to production.



Not started yet.

This chapter opens the hood on AutoGluon model stacking and shows how to read the "report card" before promoting a winning model to a KServe serving runtime.

Reading the Report Card and Shipping the Winner

Estimated reading time: 15 minutes.

Objective

Interpret AutoML evaluation artifacts and follow the production handoff path from a completed run to a served model endpoint.



Not started yet.

Under the hood: model stacking

AutoML uses AutoGluon to train and stack multiple model families — XGBoost, LightGBM, CatBoost, and FastAI/DNNs — so the leaderboard reflects an ensemble of the best performers rather than a single algorithm.

Report card analysis

Evaluate candidates using the metrics that fit the task: Accuracy for classification, R^2 for regression, and MASE for time series. Go deeper with explainability and validation tools:

- Feature Importance
- Confusion Matrices
- ROC and PR curves
- Rolling Time Series backtesting

Production handoff

Promote a winning model through a repeatable path: export the notebook, register the model in the Model Registry, and deploy to the AutoGluon ServingRuntime to expose a KServe v2 REST API.

What's Next

You can now validate and deploy a predictive model with confidence. The next chapter shifts from numbers to text, introducing generative AI in the GenAI Playground and using MLflow to govern

prompts the same way you just governed models.

Prompt Engineering and MLflow Governance

Goal

Transition from predictive data to generative text, leveraging the GenAI Playground and MLflow to version, track, and eliminate "prompt drift".



Not started yet.

This chapter bridges tabular prediction to generative text, treating prompts as versioned code and using MLflow registries to make prompt behavior auditable and repeatable.

Testing Prompts and Ending Prompt Drift

Estimated reading time: 12 minutes.

Objective

Test generative prompts interactively and use MLflow registries to version prompt templates and generation settings.



Not started yet.

From numbers to text

This section bridges tabular predictions to unstructured text generation, extending the data-to-value story from predictive models into generative AI.

The GenAI Studio Playground

Use the Playground to interactively test system instructions, context windows, and temperature settings, observing how each parameter changes model behavior before you commit to a configuration.

Prompts as code

Treat prompt templates and generation settings as versioned parameters rather than ad-hoc text. This mindset makes generative behavior reproducible and reviewable.

MLflow prompt governance

Workspace and global prompt registries solve "prompt drift" — the silent divergence of prompts over time — and provide the auditability enterprises require for generative applications.

What's Next

With prompts under version control, you are ready to ground generative models in your own data.

The next chapter introduces AutoRAG, which automatically optimizes the retrieval pipeline that feeds those prompts.

Optimizing Retrieval with AutoRAG

Goal

Position AutoRAG as an automated optimization engine that benchmarks chunking, embedding, and vector database configurations for grounded LLM applications.



Not started yet.

This chapter shows how AutoRAG grid-searches the RAG configuration space and how the diagnostic metric triad tells you whether a retrieval pipeline is production-ready.

Benchmarking the RAG Search Space

Estimated reading time: 15 minutes.

Objective

Prepare evaluation input for AutoRAG and interpret the diagnostic metric triad that scores each retrieval configuration.

▶ [autorag-demo.mp4](#) (video)

AutoRAG Demo - OpenShift AI 3.5



Not started yet.

Input preparation

AutoRAG is driven by an evaluation JSON dataset. Each test case provides a `question`, `correct_answers` (multiple valid phrasings), and `correct_answer_document_ids` that identify the grounding sources.

Automated search space and optimization

AutoRAG grid-searches the configuration space that practitioners normally tune by hand:

- Chunking strategy — recursive vs. hybrid structural
- Search mode — vector vs. hybrid
- Fusion strategy — RRF or weighted
- Embedding model — for example, BGE-M3, with multilingual auto-detection

The diagnostic metric triad

Evaluate each run against three complementary metrics:

- **Answer Faithfulness** — a hallucination check.
- **Answer Correctness** — a ground-truth match.

- **Context Correctness** — retrieval accuracy.

Production handoff and inspection

Debug pipelines visually with the Pipeline DAG, then export Responses API code snippets (cURL, Python, Go, Node.js) or indexing and inference Jupyter notebooks for handoff.

What's Next

You have now seen every capability in the course individually. The final chapter synthesizes AutoML, MLflow, and AutoRAG into one end-to-end workflow and validates your understanding with a graded assessment.

Knowledge Review and Assessment

Goal

Track core enablement concepts across AutoML, MLflow prompt tracking, and AutoRAG to validate practitioner capability in diagnosing enterprise data scenarios and selecting optimal AI patterns.



Not started yet.

This chapter connects every capability in the course into one end-to-end workflow, then validates your understanding with a scenario-based graded assessment.

Connecting Data to Models: The End-to-End Workflow

Estimated reading time: 12 minutes.

Objective

Combine AutoML, MLflow, and AutoRAG into a single decision workflow and reason about when automation alone is sufficient.



Not started yet.

End-to-end workflow synergy

The capabilities in this course combine into a live business workflow: AutoML risk scores plus AutoRAG-generated credit memos power a loan officer's decision loop, showing predictive and generative AI working together.

Diagnostic decision logic

Metric combinations tell a diagnostic story. For example, High Faithfulness combined with Low Correctness signals that the retrieval pipeline is missing document context rather than hallucinating.

When "Auto" is enough

Establish clear decision gates for staying with automated capabilities versus escalating to SLM fine-tuning (the subject of Course 2). Knowing where "Auto" stops is as important as knowing what it can do.

What's Next

With the full workflow and decision logic in view, the final section is a scenario-based graded assessment that validates your ability to diagnose enterprise data scenarios and select the optimal AI pattern.

Quiz: Diagnosing Scenarios and Selecting AI Patterns

Estimated reading time: 15 minutes.

Objective

Validate practitioner capability by diagnosing enterprise data scenarios and selecting the optimal AI pattern across AutoML, MLflow, and AutoRAG.

Interactive Assessment

```
<iframe type="text/javascript" src='_attachments/quiz.html' style="width: 1024px; height: 800px" allowfullscreen webkitallowfullscreen mozAllowFullScreen allow="autoplay *; fullscreen *; encrypted-media *" frameborder="0"></iframe>
```

What's Next

Congratulations — you have completed **Governing AI Data Workflows: AutoML and AutoRAG Strategies**. You can now strategically evaluate use cases and apply AutoML and AutoRAG in Red Hat OpenShift AI 3.5+ to drive measurable business value. This course is the first entry in the "Connecting Data to Models and Agents" series; Course 2 continues into SLM fine-tuning for scenarios where automation alone is not enough.

Appendix

FIXME: Content for Appendix...

Resources

[Download Course PDF](#)